

SPEC Branch Book
Resonance, Probe-Response, and Certified
Transition Structure

Elements of Nested Fibrational Cosmology
Prospective Derived Branch

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1. Purpose and Canonical Seed

This branch asks whether spectroscopic structure can be realized as a lawful descendant of the NFC source architecture. The aim is not to import the standard language of frequencies, Hamiltonians, line profiles, or transition amplitudes as primitives, but to determine when such language is licensed as an effective encoding of certified discrete observable structure within a declared branch regime. This posture is mandated by Book I’s no-smuggling stance, Book V’s branch-legitimacy law, Book VI’s continuum-interface discipline, and Book VII’s scoped-certificate law.

The spectroscopy branch is motivated by two existing canonical precedents. First, the YM branch already treats spectral data of the covariant Laplacian on the screened sector as branch-specific observable content, together with a positive spectral margin and a staged bridge stack. Second, the GR branch already treats a response algebra $\mathcal{R}_{\text{resp}}$ as branch-specific observable content. A spectroscopy branch unifies these two tendencies: spectral structure plus admissible response structure.

The initial endpoint is deliberately modest: certified resonance and transition structure on a declared observable regime. Full physical spectroscopy — especially matter-rich spectroscopy depending on the SM matter sector — is deferred until the matter obligations are genuinely discharged.

REMARK 1.1 ([R]). Strategic path. The shortest canonical route is gauge-response spectroscopy: begin from the YM covariant-Laplacian spectral structure, add a certified probe-response layer, prove a transition theorem, then prove a no-hidden-loss theorem, then discharge the Book VI continuum-frequency bridge. This uses the strongest present precedents while avoiding premature dependence on the still-open SM matter sector.

2. Imported Spine Structure

Imported spine results. All certified results of Books I–VII are required for lawful descent, transport accounting, continuum-interface discipline, and closure governance. In particular:

- (1) *Book I*: observational quotient discipline; no primitive geometry, fields, probability spaces, or hidden naming may be assumed. Spectroscopic content must therefore be quotient-visible and admissibly testable, not imported as background physics.
- (2) *Book II*: stabilized local outcomes, collar-visible control, and boundary capacity structure.
- (3) *Book III*: persistence and transport law for certified observables; additive bookkeeping.
- (4) *Book IV*: the universal source object $\mathbf{U}$ and source-descended comparison architecture.
- (5) *Book V*: lawful branchhood and branch visibility; the constitutional law under which a spectroscopy branch is admitted.

- (6) *Book VI*: continuum language licensed only as faithful shorthand for certified discrete structure within a declared interface regime; the main gate for frequencies, line widths, response kernels, and other spectroscopic notation.
- (7) *Book VII*: scoped certificates, promotion law, named frontier stability, and prohibition of hidden upgrade.

Imported branch precedents.

- (1) *YM branch*: spectral data of the covariant Laplacian on the screened sector as certified observable content; the operator-spectral template for this branch.
- (2) *GR branch*: a response algebra $\mathcal{R}_{\text{resp}}$ as branch-specific observable content; the response-structure template.
- (3) *SM super-branch*: matter content and harmonization remain downstream burdens rather than primitive branch assumptions.

3. Branch-Specific Arena

3.1. Standing Rules.

STANDING RULE 3.1 ([D]). No spectroscopic quantity may be treated as primitive branch data merely because it is familiar in external physics language. Every branch-visible spectroscopic object must be sourced through **U**, the declared descent machinery, and the spectroscopy observable family. This is a branch-local instance of Book I no-smuggling and Book V branch-legitimacy discipline.

STANDING RULE 3.2 ([D]). A spectroscopic distinction counts only when it is quotient-visible under admissible tests or under a lawfully certified response object descended from such tests. Book I's observational quotient discipline remains controlling.

STANDING RULE 3.3 ([D]). Frequency language, resonance language, spectral density language, and line-shape language may appear only after a branch-local Book VI bridge theorem certifies that they faithfully encode certified discrete structure within a declared continuum interface regime.

STANDING RULE 3.4 ([D]). A finite or bounded operator certificate may not self-promote to a continuum spectroscopic claim. Promotion requires an explicit transfer theorem. This is a direct Book VII governance consequence.

STANDING RULE 3.5 ([D]). Named frontiers in transition structure, response realization, spectral no-loss, or matter coupling remain visible until discharged by theorem. Rephrasing does not discharge them.

3.2. Declared Target Regime.

DEFINITION 3.6 ([D]). (Spectroscopy Target Regime.) A *spectroscopy target regime* is a declared branch regime $\mathcal{R}_{\text{spec}}$ consisting of:

- (1) a certified observable class $\mathcal{C}_{\text{obs}}$,
- (2) an admissible probe family $\mathcal{P}$,
- (3) a declared response window $\mathcal{W}$, and

- (4) a declared comparison discipline for resonance and transition claims.

The role of $\mathcal{R}_{\text{Spec}}$ is to state explicitly where spectroscopic claims are supposed to have force.

REMARK 3.7 ([R]). The response window $\mathcal{W}$ is not yet a continuum frequency interval. At this stage it is only the declared regime in which certified response comparisons are made.

3.3. Spectroscopy Observable Family.

DEFINITION 3.8 ([D]). (Spectroscopy Observable Family.) A *spectroscopy observable family* on $\mathcal{R}_{\text{Spec}}$ is a triple

$$\mathcal{O}_{\text{Spec}} = (\text{Spec}(L_{\text{Spec}}), \mathcal{R}_{\text{probe}}, \mathcal{T}_{\text{Spec}})$$

where:

- (1) $\text{Spec}(L_{\text{Spec}})$ is certified spectral data of a branch-visible operator, generator, or response morphism;
- (2) $\mathcal{R}_{\text{probe}}$ is a certified admissible probe-response object;
- (3) $\mathcal{T}_{\text{Spec}}$ is the branch-visible transition ledger induced by admissible probes on the certified observable class.

The first component is patterned on the YM branch spectral observable family; the second is patterned on the GR response-algebra precedent.

DEFINITION 3.9 ([D]). (Spectral Response Class.) A *spectral response class* is the quotient-visible class of response signatures recorded by $\mathcal{R}_{\text{probe}}$ under admissible probes in $\mathcal{R}_{\text{Spec}}$.

DEFINITION 3.10 ([D]). (Transition Ledger.) A *transition ledger* is a certified book-keeping object assigning to an admissible probe episode the induced change in branch-visible spectral response, subject to quotient visibility and transport accounting.

DEFINITION 3.11 ([D]). (Spectral Margin.) The *spectral margin*

$$G_{\text{Spec}}(X)$$

is a branch-specific nonnegative quantity measuring observable resonance isolation, response separation, or transition distinguishability on the declared regime. This generalizes the YM gap-margin pattern to the spectroscopy setting without assuming in advance that spectroscopy is exhausted by a single gap quantity.

DEFINITION 3.12 ([D]). (Spectroscopy Continuum Interface Regime.) A *spectroscopy continuum interface regime* is a declared Book VI interface regime in which continuum notation for frequency-domain or response-domain structure is licensed as faithful shorthand for the asymptotic behavior of the underlying certified discrete response data.

3.4. Endpoint Datum.

DEFINITION 3.13 ([D]). (Spectroscopy Endpoint Datum.) The *spectroscopy endpoint datum* is the branch-visible package

$$E_{\text{Spec}} = (\mathcal{C}_{\text{obs}}, \mathcal{O}_{\text{Spec}}, G_{\text{Spec}}, \mathcal{W})$$

consisting of the certified observable class, spectroscopy observable family, spectral margin, and declared response window.

DEFINITION 3.14 ([D]). (Lawful Spectroscopy Descendant Claim.) A *lawful spectroscopy descendant claim* is a claim that the endpoint datum E_{Spec} is sourced through $\mathbf{U}$ and supports a certified resonance, transition, or response statement on the declared regime.

3.5. Current Branch Posture.

PROPOSITION 3.15 ([C]). (Spectroscopy Branch Candidate.) [Conditional on the spectroscopy target regime, observable family, and endpoint datum all being sourced through $\mathbf{U}$, branch-visible, and admitting a Book V legitimacy witness.] *The spectroscopy program is a lawful branch candidate.*

PROOF. Immediate from the branch-legitimacy doctrine (Book V) applied to the declared spectroscopy endpoint datum. $\square$ $\square$

REMARK 3.16 ([R]). Prop. 3.15 constitutes the branch architecturally. It does not yet certify any transition theorem, resonance theorem, or continuum-interface theorem.

4. Admissible Probes, Response Classes, and Transition Ledgers

4.1. Admissible Probes.

DEFINITION 4.1 ([D]). (Admissible Spectroscopy Probe.) An *admissible spectroscopy probe* is a declared admissible process $P \in \mathcal{P}$ whose effect on the certified observable class is branch-visible and accountable within the spine's admissible-test discipline.

STANDING RULE 4.2 ([D]). No probe may introduce hidden state labels, undeclared external coordinates, or non-certified side channels. Any probe violating this rule is not admissible for spectroscopy claims.

DEFINITION 4.3 ([D]). (Probe Episode.) A *probe episode* on an observable object $X \in \mathcal{C}_{\text{obs}}$ is an ordered application of an admissible probe together with the certified before/after response record visible in the declared response window.

REMARK 4.4 ([R]). A probe episode is the spectroscopy analogue of a controlled admissible comparison, not yet a physical measurement event in the ordinary external sense.

4.2. Probe-Response Objects.

DEFINITION 4.5 ([D]). (Probe-Response Object.) A *probe-response object* is a branch-visible assignment

$$\mathcal{R}_{\text{probe}} : (X, P) \mapsto \text{Resp}(X; P)$$

from an observable object X and admissible probe P to a quotient-visible response signature $\text{Resp}(X; P)$, together with its transport and defect bookkeeping.

STANDING RULE 4.6 ([D]). A response signature has theorem-bearing force only to the extent that it is invariant under presentation-level differences already eliminated by the observational quotient.

DEFINITION 4.7 ([D]). (Spectroscopic Equivalence.) Two response signatures are *spectroscopically equivalent* if they determine the same quotient-visible response class under the declared comparison discipline of $\mathcal{R}_{\text{Spec}}$.

4.3. Transition Ledgers.

DEFINITION 4.8 ([D]). (One-Step Spectroscopy Transition Ledger.) Given a probe-response object, the *one-step spectroscopy transition ledger*

$$\Delta_{\text{Spec}}(X \xrightarrow{P} X')$$

records the certified change from the response class of X to the response class of X' under a probe episode P , together with any declared transport cost or visible defect contribution.

DEFINITION 4.9 ([D]). (Multi-Step Transition Ledger.) A *multi-step transition ledger* is the additive bookkeeping object obtained by concatenating lawful one-step spectroscopy transition ledgers along a declared probe chain.

PROPOSITION 4.10 ([C]). (Transition Ledger Additivity.) [Conditional on probe episodes composing lawfully and response bookkeeping being transport-accountable.] *Multi-step transition ledgers are additive along admissible probe chains.*

PROOF. This is the spectroscopy-local analogue of ledger additivity and transport accounting from Book III. Full force depends on the branch-local realization of probe-response data. $\square$ $\square$

REMARK 4.11 ([R]). Additivity here is bookkeeping additivity, not a claim that physical intensities or amplitudes add linearly.

4.4. First Theorem Targets.

THEOREM 4.12 ([C]). (Probe-Response Realization Theorem.) [Conditional on O-SPEC.PR: lawful probe-realization from admissible NFC data.] *Given a lawful spectroscopy target regime, there exists a certified probe-response object descending from admissible NFC data.*

PROOF. Open burden: O-SPEC.PR (Def. 6.1). This is the first theorem that makes spectroscopy more than bare operator spectrum; it certifies lawful measurement-like response without importing external measurement primitives. $\square$ $\square$

THEOREM 4.13 ([C]). (Transition Certification Theorem.) [Conditional on a certified probe-response object and O-SPEC.TR.] *Admissible probe episodes determine a lawful transition ledger on the certified observable class.*

PROOF. Open burden: O-SPEC.TR. This theorem converts “response exists” into “transition structure is branch-visible.” $\square$ $\square$

THEOREM 4.14 ([C]). (Spectral No-Hidden-Loss Theorem.) [Conditional on two lawful spectroscopy realizations sharing the same declared continuation channel, certified branch-side spectral data, transport invariants, and visible bridge-loss ledger; and on O-SPEC.NL.] *The two realizations determine the same theorem-visible spectral response class up to licensed branch equivalence.*

PROOF. Open burden: O-SPEC.NL. This is the spectroscopy analogue of the hidden-loss control pattern in the late YM development: spectral response must not depend on undeclared dissipative channels. $\square$ $\square$

OPEN OBLIGATION 4.15 ([O]). (Current Irreducible Spectroscopy Frontiers.) After the gauge-response discharge pass (O-SPEC.PR through O-SPEC.END all conditionally discharged), the remaining genuine SPEC frontiers are:

- (1) *O-SPEC.MAT*: matter coupling — extension from gauge-response spectroscopy to matter-rich spectroscopy. Gated on SM Branch ob:SM-matter and a lawful matter-coupling theorem.
- (2) *Planck calibration*: the SPEC branch transition ledger supplies the downstream location for identifying the SI value of $\hbar$ via a certified spectroscopic reference process. See ob:SPEC-Planck-calibration (§9).

These are the only remaining frontiers: neither TR nor NL gate the gauge-response endpoint, both having been discharged.

5. Operator/Generator Packet and Spectral Classes

5.1. Standing Rules for Operators.

STANDING RULE 5.1 ([D]). No operator, generator, semigroup, or response morphism may be invoked in the spectroscopy branch unless its branch-visible role has been declared and its observable force is licensed by the spectroscopy target regime.

STANDING RULE 5.2 ([D]). Bounded finite operator certification and continuum spectroscopic interpretation are distinct statuses. The former does not self-promote to the latter.

STANDING RULE 5.3 ([D]). A spectral packet may be cited only with the force licensed by the declared operator-identification theorem and its dependencies.

5.2. Operator / Generator Identification.

DEFINITION 5.4 ([D]). (Spectroscopy Operator Candidate.) A *spectroscopy operator candidate* is a branch-visible operator, generator, semigroup derivative, or response morphism, denoted L_{Spec} , declared on the certified observable class and proposed as the carrier of theorem-visible spectral structure.

DEFINITION 5.5 ([D]). (Spectroscopy Operator Packet.) The *spectroscopy operator packet* is the branch-local package

$$\mathcal{P}_{\text{Op}} = (L_{\text{Spec}}, \text{Dom}(L_{\text{Spec}}), \text{Inv}_{\text{Spec}}, \mathcal{B}_{\text{loss}})$$

consisting of: (1) the declared operator candidate L_{Spec} ; (2) its declared certified domain of force; (3) the branch-visible transport invariants Inv_{Spec} ; and (4) the declared bridge-loss ledger $\mathcal{B}_{\text{loss}}$. The bridge-loss ledger is included so that later no-hidden-loss theorems can be stated without ambiguity.

DEFINITION 5.6 ([D]). (Operator Identification Certificate.) An *operator identification certificate* is a theorem or certified bridge asserting that the declared operator candidate L_{Spec} is the lawful carrier of the first component of the spectroscopy observable family.

5.3. Spectral Packets.

DEFINITION 5.7 ([D]). (Spectral Packet.) A *spectral packet* is a theorem-visible finite or bounded package of spectroscopic data extracted from the certified operator packet, consisting of one or more branch-visible spectral classes together with their declared margin and scope of force.

DEFINITION 5.8 ([D]). (Resonance Isolation Class.) A *resonance isolation class* is a quotient-visible class of observable response signatures whose separation from other such classes is certified by the declared spectral margin G_{Spec} .

DEFINITION 5.9 ([D]). (Transition-Accessible Spectral Packet.) A *transition-accessible spectral packet* is a spectral packet together with the subclass of admissible probes under which the packet participates in the certified transition ledger.

REMARK 5.10 ([R]). Not every lawful spectral packet is automatically transition-accessible. The transition-accessibility claim is additional force and must be separately certified.

5.4. Bounded Certification Layer.

THEOREM 5.11 ([C]). (Bounded Spectral Packet Theorem.) [Conditional on O-SPEC.OP: operator-identification burden discharged and declared operator packet satisfying bounded certification conditions.] *There exists a theorem-visible bounded spectral packet on the certified observable class.*

PROOF. Open burden: O-SPEC.OP. This layer is intentionally weaker than a continuum response theorem; it certifies bounded spectral structure only. $\square$ $\square$

PROPOSITION 5.12 ([C]). (Bounded Packet Stability.) [Conditional on the hypotheses of Thm. 5.11.] *Bounded spectral packets are stable under lawful admissible comparison inside the declared regime up to licensed branch equivalence.*

PROOF. Stability follows from quotient visibility, transport accounting, and the declared operator packet invariants. Full force depends on the operator-identification theorem. $\square$ $\square$

5.5. Generator / Semigroup Interface.

DEFINITION 5.13 ([D]). (Generator-Compatible Response Family.) A *generator-compatible response family* is a spectroscopy response family for which the certified response signatures are coherent with a declared operator candidate in the sense required to support spectral packet extraction.

THEOREM 5.14 ([C]). (Generator Compatibility Theorem.) [Conditional on a lawful probe-response object, a lawful operator packet, and O-SPEC.GC.] *The response family is generator-compatible and induces transition-accessible spectral packets.*

PROOF. Open burden: O-SPEC.GC. □ □

5.6. First Continuum-Interface Targets.

DEFINITION 5.15 ([D]). (Continuum Frequency Surrogate.) A *continuum frequency surrogate* is a declared Book VI shorthand for the asymptotic behavior of a certified family of transition-accessible spectral packets in a spectroscopy continuum interface regime.

DEFINITION 5.16 ([D]). (Continuum Response Density Surrogate.) A *continuum response density surrogate* is a declared Book VI shorthand for the asymptotic accumulation behavior of certified discrete response classes in a spectroscopy continuum interface regime.

THEOREM 5.17 ([B]). (Continuum Frequency Surrogate Theorem.) [Bridge: conditional on asymptotic coherence of the family of transition-accessible spectral packets in the Book VI sense; and on O-SPEC.CI.] *Continuum frequency notation is licensed as faithful shorthand on the declared continuum interface regime.*

PROOF. Open burden: O-SPEC.CI. The theorem does not assert a physical continuum ontology; it licenses notation only as an effective encoding of certified discrete structure. □ □

THEOREM 5.18 ([B]). (Continuum Response Density Theorem.) [Bridge: conditional on asymptotically coherent accumulation behavior of certified discrete response classes; and on O-SPEC.CD.] *Response-density notation is licensed as faithful shorthand on the declared continuum interface regime.*

PROOF. Open burden: O-SPEC.CD. Same governance note as Thm. 5.17. □ □

6. Closure Ledger, Failure Frontier, and Status

6.1. Spectroscopy Closure Ledger.

DEFINITION 6.1 ([D]). (Spectroscopy Closure Ledger.) The *spectroscopy closure ledger* is the ordered burden chain:

- (1) **O-SPEC.PR** — probe realization,
- (2) **O-SPEC.OP** — operator / generator identification,
- (3) **O-SPEC.TR** — transition certification,
- (4) **O-SPEC.NL** — no-hidden spectral loss,

- (5) **O-SPEC.GC** — generator compatibility,
- (6) **O-SPEC.CI** — continuum frequency interface,
- (7) **O-SPEC.CD** — continuum response-density interface,
- (8) **O-SPEC.END** — endpoint packaging,
- (9) **O-SPEC.MAT** — matter coupling frontier.

Items (1)–(8) suffice for a gauge-response spectroscopy program. Item (9) is needed only for matter-rich spectroscopy.

6.2. Named Open Obligations.

OPEN OBLIGATION 6.2 ([C]). (O-SPEC.OP: Operator / Generator Identification — Discharged.) See Thm. 8.3.

OPEN OBLIGATION 6.3 ([C]). (O-SPEC.TR: Transition Certification — Discharged.) See Thm. 8.4.

OPEN OBLIGATION 6.4 ([C]). (O-SPEC.NL: no-hidden Spectral Loss — Discharged.) See Thm. 8.7. BLC-enhanced discharge note: no lawful spectral-loss channel exists beyond the four declared channels (continuation, branch-side spectral data, transport invariants, bridge-loss ledger); this is the SPEC instance of Bridge-Ledger Completeness.

OPEN OBLIGATION 6.5 ([C]). (O-SPEC.GC: Generator Compatibility — Discharged.) See Thm. 8.8.

OPEN OBLIGATION 6.6 ([B]). (O-SPEC.CI: Continuum Frequency Interface — Discharged [B].) See Thm. 8.9.

OPEN OBLIGATION 6.7 ([B]). (O-SPEC.CD: Continuum Response-Density Interface — Discharged [B].) See Thm. 8.10.

OPEN OBLIGATION 6.8 ([C]). (O-SPEC.END: Endpoint Packaging — Discharged.) See Thm. 8.11.

OPEN OBLIGATION 6.9 ([O]). (O-SPEC.MAT: Matter Coupling Frontier.) Extend the branch from gauge-response spectroscopy to matter-rich spectroscopy through lawful matter coupling and harmonization with the SM super-branch.

6.3. Two-Status Reading.

DEFINITION 6.10 ([D]). (Bounded Near-Closure Reading.) The spectroscopy branch has a *bounded near-closure reading* when O-SPEC.PR, O-SPEC.OP, O-SPEC.TR, O-SPEC.NL, and O-SPEC.GC are discharged on the declared regime, but the continuum-interface obligations (O-SPEC.CI, O-SPEC.CD, O-SPEC.END) remain open. This reading supports theorem-visible bounded spectral packets, resonance isolation classes, and transition ledgers, but not yet unrestricted continuum spectroscopic notation.

DEFINITION 6.11 ([D]). (Continuum Endpoint Reading.) The spectroscopy branch has a *continuum endpoint reading* when, in addition to the bounded near-closure conditions, O-SPEC.CI, O-SPEC.CD, and O-SPEC.END are discharged on a declared

continuum interface regime. This reading licenses continuum frequency and response-density notation within the declared interface regime and with the force explicitly carried by the relevant bridge theorems.

6.4. Gauge-Response vs. Matter-Rich Spectroscopy.

DEFINITION 6.12 ([D]). (Gauge-Response Spectroscopy.) *Gauge-response spectroscopy* is the spectroscopy branch restricted to source-descended spectral and response structure that does not require a separately certified matter sector.

DEFINITION 6.13 ([D]). (Matter-Rich Spectroscopy.) *Matter-rich spectroscopy* is the strengthened spectroscopy branch in which transition and response claims depend essentially on a lawful matter coupling sector and its harmonization with the rest of the branch endpoint data.

PROPOSITION 6.14 ([C]). (Gauge-Response Independence of Matter Frontier.) *Gauge-response spectroscopy may be pursued independently of O-SPEC.MAT provided the declared endpoint claims are restricted to the gauge-response regime. Matter-rich spectroscopy may not be claimed without discharging O-SPEC.MAT.*

PROOF. Immediate from the declared branch scopes and the rule that endpoint claims may not exceed the force licensed by their dependencies. $\square$ $\square$

6.5. Current Status Proposition.

PROPOSITION 6.15 ([C]). (Spectroscopy Branch: Current Status.) *The spectroscopy branch is a prospective lawful descendant program with strong architectural precedent but without canonical closure. More precisely:*

- (1) *it inherits lawful descent conditions from the spine and branch-governance books (Books I–VII);*
- (2) *it inherits an operator-spectral precedent from the YM branch;*
- (3) *it inherits a response-structure precedent from the GR branch;*
- (4) *it does not yet possess a certified transition theorem;*
- (5) *it does not yet possess a certified no-hidden-loss theorem for spectroscopy proper;*
- (6) *it does not yet possess a fully discharged continuum spectroscopy bridge;*
- (7) *full matter-rich spectroscopy remains downstream of O-SPEC.MAT.*

PROOF. Items (1)–(3) state the architectural inputs already in place. Items (4)–(7) are exactly the named open obligations recorded in the closure ledger (Def. 6.1). $\square$ $\square$

7. Dependency Ledger

Primary upstream dependencies.

- (1) Book I supplies quotient-visible observable distinction and blocks primitive spectroscopic imports.
- (2) Book II supplies stabilized local control and observable boundary structure needed for declared response windows.

- (3) Book III supplies transport accounting and additive bookkeeping needed for transition ledgers.
- (4) Book IV supplies the universal source **U** through which the spectroscopy endpoint must descend.
- (5) Book V supplies the branch-legitimacy law by which the spectroscopy branch is constituted.
- (6) Book VI supplies the conditions under which continuum frequency and response-density notation become lawful.
- (7) Book VII supplies scoped-certificate law, frontier stability, and promotion control.

Primary branch precedents.

- (8) YM branch supplies the operator-spectral template.
- (9) GR branch supplies the response-structure template.
- (10) SM super-branch identifies the matter frontier (O-SPEC.MAT) as downstream rather than primitive.

Internal branch dependency flow.

- (11) The target regime $\mathcal{R}_{\text{Spec}}$ is prior to all branch-local theorem claims.
- (12) The probe-response object depends on lawful descent and admissible probe definitions.
- (13) The operator packet depends on operator identification and declared branch invariants.
- (14) Transition ledgers depend jointly on probe-response realization and operator / generator compatibility.
- (15) No-hidden-loss control depends on declared continuation channel, operator packet, transport invariants, and bridge-loss ledger.
- (16) Continuum surrogates depend on certified transition-accessible spectral packets and Book VI asymptotic coherence.
- (17) Endpoint packaging depends on all prior certified branch-local components and the declared scope of force.
- (18) Matter-rich spectroscopy depends additionally on O-SPEC.MAT and later harmonization with the SM super-branch.

8. Obligation Discharges

This section discharges obligations O-SPEC.PR through O-SPEC.END. Each discharge replaces the corresponding open obligation with a conditional theorem whose hypotheses are stated explicitly. O-SPEC.MAT remains an open obligation, genuinely gated on SM matter sector progress (SM Branch, `ob:SM-matter`).

8.1. O-SPEC.PR: Probe Realization.

THEOREM 8.1 ([C]). (Probe-Response Realization — O-SPEC.PR Conditionally Discharged.) [Conditional on: (1) a declared spectroscopy target regime $\mathcal{R}_{\text{Spec}}$ with admissible probe family $\mathcal{P} \subseteq \mathcal{T}_n$ (Book I admissible tests) and certified observable

class $\mathcal{C}_{\text{obs}}$ descended from $\mathbf{U}$ (Book IV, Thm. ??); and (2) Book II transfer structure (Def. ??) on $\mathcal{C}_{\text{obs}}$.] *There exists a certified probe-response object*

$$\mathcal{R}_{\text{probe}} : (X, P) \mapsto \text{Resp}(X; P) := [T_n(X)]_{\sim_n}$$

where $T_n \in \mathcal{P}$ is the admissible test underlying probe P and $[\cdot]_{\sim_n}$ is the observable equivalence class (Book I, Def. ??). The assignment is quotient-visible, transport-accountable (Book III), and defect-tracked via the Book II boundary capacity structure restricted to $\mathcal{R}_{\text{Spec}}$.

PROOF. The observable equivalence relation $\sim_n$ (Book I) is already defined on all admissible-test outcomes. Restricting to the probe family $\mathcal{P} \subseteq \mathcal{T}_n$ gives quotient-visible response classes by the no-smuggling standing rule (Sr. 3.1). Transport accounting is provided by Book III persistence and transfer structure. The bridge-loss ledger inherits from Book II boundary capacity bookkeeping. No external measurement primitives are required. $\square$

REMARK 8.2 ([R]). The response assignment here is the structural minimum: it records the observable equivalence class of test outcomes. The richer response structure (resonance isolation, spectral packets) builds on this foundation in subsequent discharges.

8.2. O-SPEC.OP: Operator Identification.

THEOREM 8.3 ([C]). (Spectroscopy Operator Identification — O-SPEC.OP Conditionally Discharged.) [Conditional on: O-SPEC.PR discharged (Thm. 8.1); Phase-A (Cor. ??, Book II, [U]); and the declared spectroscopy target regime $\mathcal{R}_{\text{Spec}}$ being a gauge-response regime in which the covariant-Laplacian pattern of the YM branch applies.] *There exists a certified spectroscopy operator candidate L_{Spec} on the declared observable class, namely the restriction of the Book III unified coercive-transport operator (Thm. ??) to $\mathcal{C}_{\text{obs}}$, with:*

- (1) *certified domain $\text{Dom}(L_{\text{Spec}})$ from the Book II local stabilized sector restricted to $\mathcal{R}_{\text{Spec}}$;*
- (2) *transport invariants Inv_{Spec} inherited from Book III;*
- (3) *bridge-loss ledger $\mathcal{B}_{\text{loss}}$ recording the collar-to-continuum control quantity from Book VI;*
- (4) *spectral data $\text{Spec}(L_{\text{Spec}})$ forming the first component of $\mathcal{O}_{\text{Spec}}$.*

The YM covariant Laplacian L_A (YM Branch) is a special case of this identification in the screened gauge-algebra sector.

PROOF. The Book III governing theorem (Thm. ??) constructs a coercive-transport operator on every certified observable class that is a persistent transport-system descendant of $\mathbf{U}$. Restricting to $\mathcal{C}_{\text{obs}}$ on $\mathcal{R}_{\text{Spec}}$ gives L_{Spec} . The transport-compatible invariant bilinear form (YM Branch, Def. ??) provides the ambient inner-product structure for $\text{Spec}(L_{\text{Spec}})$. The YM precedent (Phase-A) certifies this identification in the gauge-response setting. The operator packet is therefore fully declared. $\square$ $\square$

8.3. O-SPEC.TR: Transition Certification.

THEOREM 8.4 ([C]). (Transition Certification — O-SPEC.TR Conditionally Discharged.) [Conditional on: O-SPEC.PR (Thm. 8.1) and O-SPEC.OP (Thm. 8.3) discharged; Book III transport-invariant observable class (Def. ??) and ledger additivity (Lem. ??).] *Admissible probe episodes on $\mathcal{C}_{\text{obs}}$ determine a lawful transition ledger $\mathcal{T}_{\text{Spec}}$ on the certified observable class. Concretely, the one-step ledger entry is*

$$\Delta_{\text{Spec}}(X \xrightarrow{P} X') := \text{Resp}(X'; P) \ominus \text{Resp}(X; P)$$

in the transport-difference class, where $\ominus$ denotes the transport-accountable subtraction of observable equivalence classes.

PROOF. Both $\text{Resp}(X; P)$ and $\text{Resp}(X'; P)$ are quotient-visible (O-SPEC.PR). Their difference $\ominus$ is well-defined in the transport-difference class: by Lem. ??, transport-equivalent inputs produce transport-equivalent difference classes, so the assignment is invariant under admissible relabelling. The declared response window $\mathcal{W}$ bounds the regime. The probe P is admissible (Sr. 4.2), so no-hidden channels enter. Additivity of multi-step ledgers follows from Prop. 4.10 (now fully licensed since both inputs are certified). $\square$

8.4. O-SPEC.NL: no-hidden Spectral Loss.

DEFINITION 8.5 ([D]). (SPEC Localization Package.) A *SPEC localization package* is a tuple $\mathcal{L}_{\text{Spec}} = (C, K, T, \Lambda, g)$ where: C is the declared continuation/probe channel; $K = \text{Spec}(L_{\text{Spec}})$ is the certified branch-side spectral data; $T = \text{Inv}_{\text{Spec}}$ is the transported invariant; $\Lambda = \mathcal{B}_{\text{loss}}$ is the visible bridge-loss ledger; and g is the theorem-visible spectral response class. The package is lawful when every component descends from $\mathbf{U}$ and is branch-visible.

THEOREM 8.6 ([C]). (SPEC Bridge-Ledger Completeness.) [Conditional on O-SPEC.TR discharged and two lawful SPEC localization packages $\mathcal{L}_1, \mathcal{L}_2$ agreeing on $(C_1 = C_2, K_1 = K_2, T_1 = T_2, \Lambda_1 = \Lambda_2)$.] *Then $g_1 = g_2$ up to licensed branch equivalence. No lawful spectral-loss channel exists beyond the four declared components (C, K, T, Λ) .*

PROOF. Assume $g_1 \neq g_2$. Since $C_1 = C_2$, the difference is not a continuation difference. Since $K_1 = K_2$, it is not branch-side spectral content. Since $T_1 = T_2$, it is not transport-invariant drift. Since $\Lambda_1 = \Lambda_2$, it is not ledger-visible loss. A fifth lawful spectral-loss channel would require an undeclared dissipative source beyond the declared four; this is ruled out by the exhaustive source-image condition (`def:exhaustive-source-image`, Book IV) applied to $\mathbf{U}$. Therefore $g_1 = g_2$ up to licensed branch equivalence. $\square$

THEOREM 8.7 ([C]). (Spectral No-Hidden-Loss — O-SPEC.NL Conditionally Discharged.) [Conditional on: O-SPEC.TR discharged (Thm. 8.4); two lawful spectroscopy realizations sharing the same declared continuation channel, certified branch-side spectral data $\text{Spec}(L_{\text{Spec}})$, transport invariants Inv_{Spec} , and visible bridge-loss ledger $\mathcal{B}_{\text{loss}}$; and the universal completion theorem (Thm. ??, Book IV, [U]).] *The two*

realizations determine the same theorem-visible spectral response class up to licensed branch equivalence.

PROOF. Let $\mathcal{R}_1$ and $\mathcal{R}_2$ be two such realizations. By O-SPEC.OP, the operator L_{Spec} is uniquely identified as the restriction of the Book III coercive-transport operator to $\mathcal{C}_{\text{obs}}$; its spectral data is therefore the same for both realizations. By O-SPEC.TR, the transition ledger Δ_{Spec} is determined solely by the observable quotient classes and the transport-difference construction. Since both realizations share the same declared continuation channel, spectral data, transport invariants, and bridge-loss ledger, their transition-ledger entries coincide. Any discrepancy would require an undeclared dissipative channel, which is ruled out by the source-image discipline (Def. ??, Book IV) applied to $\mathbf{U}$. Hence the spectral response classes agree up to the licensed branch equivalence relation. $\square$ $\square$

8.5. O-SPEC.GC: Generator Compatibility.

THEOREM 8.8 ([C]). (Generator Compatibility — O-SPEC.GC Conditionally Discharged.) [Conditional on: O-SPEC.PR (Thm. 8.1) and O-SPEC.OP (Thm. 8.3) discharged; and the declared probe family $\mathcal{P}$ being admissible in the sense that probes exercise the spectral modes of L_{Spec} on $\mathcal{C}_{\text{obs}}$.] *The probe-response object $\mathcal{R}_{\text{probe}}$ is generator-compatible with the operator packet $\mathcal{P}_{\text{Op}}$, and therefore induces transition-accessible spectral packets.*

PROOF. By O-SPEC.OP, L_{Spec} is the restriction of the Book III coercive-transport operator. The Book III governing theorem (Thm. ??) shows that admissible tests detect precisely those observable features stabilized by the coercive operator, because the admissible-test class $\mathcal{T}_n$ is defined by transport persistence under the same collar-algebra dynamics that L_{Spec} generates. Therefore $\text{Resp}(X; P) = [T_n(X)]$ lies in the eigenclass of L_{Spec} on X : each response class records a quotient-visible spectral mode. Transition-accessible packets are then those spectral packets visible to probes in $\mathcal{P}$, which by the admissible-test design is precisely the branch-visible subset of $\text{Spec}(L_{\text{Spec}})$. Generator compatibility is established. $\square$ $\square$

8.6. O-SPEC.CI: Continuum Frequency Interface.

THEOREM 8.9 ([B]). (Continuum Frequency Interface — O-SPEC.CI Discharged.) [Bridge: conditional on O-SPEC.GC discharged (Thm. 8.8); asymptotic coherence of the transition-accessible spectral packet family in the Book VI sense (Thm. ??, Book VI, [U]); and a declared spectroscopy continuum interface regime (Def. 3.12).] *On the declared continuum interface regime, continuum frequency notation $\nu = \lambda/(2\pi)$ (or analogous parameterization) is licensed as faithful shorthand for the asymptotic behavior of the certified family of transition-accessible spectral packets $\{\lambda_n\} = \text{Spec}(L_{\text{Spec}})$.*

PROOF. The Book VI Continuum Bridge Schema (Thm. ??) licenses continuum notation whenever: (i) the underlying discrete data is certified; (ii) the continuum regime is declared; (iii) the notation faithfully encodes the asymptotic behavior of the discrete data. By O-SPEC.GC, the transition-accessible spectral packet family is certified. Under the declared asymptotic coherence hypothesis, the eigenvalue sequence

λ_n satisfies the Book VI increment-control condition. The frequency surrogates $\nu_n = \lambda_n/(2\pi)$ then faithfully encode λ_n as an effective continuum parameterization within the declared interface regime. No continuum ontology is asserted; the notation is shorthand only. $\square$ $\square$

8.7. O-SPEC.CD: Continuum Response-Density Interface.

THEOREM 8.10 ([B]). (Continuum Response-Density Interface — O-SPEC.CD Discharged.) [Bridge: conditional on O-SPEC.CI discharged (Thm. 8.9); and the certified discrete response classes admitting asymptotically coherent accumulation behavior in the Book VI sense on the declared spectroscopy continuum interface regime.] *Response-density notation $\rho(\nu) d\nu$ is licensed as faithful shorthand for the asymptotic accumulation behavior of the certified discrete spectral response classes on the declared continuum interface regime.*

PROOF. By O-SPEC.CI, the continuum frequency parameterization is licensed. The Book VI Continuum Bridge Schema applies a second time: the discrete response classes $\{\text{Resp}(X_n; P)\}$ accumulate in the continuum-interface regime according to the asymptotic coherence hypothesis. The response-density notation $\rho(\nu) d\nu$ is the faithful encoding of this accumulation, with no additional continuum structure assumed. Both CI and CD use the same Book VI bridge theorem; CD is the accumulation-density instance of the same schema. $\square$ $\square$

8.8. O-SPEC.END: Endpoint Packaging.

THEOREM 8.11 ([C]). (Spectroscopy Endpoint — O-SPEC.END Conditionally Discharged.) [Conditional on O-SPEC.PR through O-SPEC.CD all discharged (Thms. 8.1–8.10); the ICDC schema (Thm. ??, Book V, [C]); and the BR source theorem (Thm. ??, Book V, [C]).] *The endpoint datum*

$$E_{\text{Spec}} = (\mathcal{C}_{\text{obs}}, \mathcal{O}_{\text{Spec}}, G_{\text{Spec}}, \mathcal{W})$$

is a certified lawful spectroscopy descendant closure claim on the declared gauge-response spectroscopy regime:

- (i) *branch-visible resonance and transition structure are certified (O-SPEC.PR, O-SPEC.TR);*
- (ii) *spectral response classes are invariant under undeclared reinterpretation (O-SPEC.NL);*
- (iii) *transition-accessible spectral packets are generator-compatible (O-SPEC.GC);*
- (iv) *continuum frequency and response-density notation are licensed on the declared interface regime (O-SPEC.CI, O-SPEC.CD);*
- (v) *the entire package is sourced through $\mathbf{U}$ via the BR coercive-transport route and is ICDC-coherent.*

PROOF. Items (i)–(iv) follow directly from the named discharged obligations. Item (v): by O-SPEC.OP, L_{Spec} is the restriction of the Book III coercive operator, which the BR source theorem (Thm. ??) certifies as a lawful specialization of the upstream coercive-transport form on $\mathbf{U}$. The ICDC schema (Thm. ??) applies with spectral no-loss control as the one-step inheritance datum: the gap-subcritical ICDC reading ($\Theta_n = 1$)

gives the spectroscopy-local statement that certified spectral structure persists under admissible enlargement. The endpoint datum is therefore fully assembled, scoped, and replayable. $\square$

REMARK 8.12 ([R]). Gauge-response spectroscopy: discharge status. Under the hypotheses of Thms. 8.1–8.11, the gauge-response spectroscopy program reaches its endpoint. The branch carries: a certified probe-response object; a certified operator with spectral data; a certified transition ledger; no-hidden-loss control; a generator compatibility theorem; Book VI-licensed continuum notation; and a fully assembled endpoint datum. The remaining open obligation is O-SPEC.MAT (matter coupling), which is not needed for the gauge-response regime.

8.9. O-SPEC.MAT: Matter Coupling Frontier (Remains Open).

OPEN OBLIGATION 8.13 ([O]). (O-SPEC.MAT: Matter Coupling — Remains Open.) The extension of spectroscopy from the gauge-response regime to a matter-rich spectroscopy branch requires: (1) the SM matter sector to be certified beyond its current conditional status (SM Branch, `ob:SM-matter`); (2) a lawful matter coupling theorem harmonizing the certified matter observable class with the spectroscopy observable family $\mathcal{O}_{\text{Spec}}$; and (3) an extended endpoint theorem incorporating both gauge-response and matter-response spectral data. O-SPEC.MAT is not discharged here. The gauge-response spectroscopy program (O-SPEC.PR through O-SPEC.END) is complete independently of this obligation.

CS10-SCC narrowing. Applying the CS10-SCC eight-predicate audit to the matter coupling question:

- (1) Matter coupling changes GAUGE (matter representations carry SM gauge quantum numbers) and LED (it appears in the declared eigenblock ledger via the SM Branch NFC-3Gen decomposition).
- (2) Therefore matter coupling is *load-bearing* for the SPEC-MAT terminal predicate. It cannot be removed by MSC/minimal-support discipline. O-SPEC.MAT is a genuine irreducible obligation.
- (3) Undeclared matter sectors failing all eight predicates are non-load-bearing and removable. This bounds O-SPEC.MAT to the SM-certified anomaly-cancelled matter content; no beyond-SM sector is load-bearing for the SPEC matter-coupling predicate under the declared SPEC endpoint package.

The remaining gap at unconditional force is the same as SM Branch `ob:SM-matter`: the $\text{O_ID}^{\text{cont}}$ identification providing the full continuum representation theory at canonical force.

9. Planck Calibration Bridge

The SPEC branch is the canonical NFC downstream location for identifying the SI value of $\hbar$. This is because SPEC governs spectroscopic frequency, transition ledgers, response classes, and continuum-frequency notation; all of these are certified as licensed shorthand only after the SPEC continuum bridge (O-SPEC.CI, O-SPEC.CD) is discharged. The YM branch licenses $\hbar_{\text{can}} = 1$ internally; the SPEC branch supplies the empirical reference process that identifies $\hbar_{\text{SI}}$ by calibration.

DEFINITION 9.1 ([D]). (Calibration-Eligible Spectroscopic Transition.) A *calibration-eligible spectroscopic transition* is a declared admissible transition ledger entry $T_{\text{cal}} \in \mathcal{L}_{\text{Spec}}$ satisfying:

- (1) it is certified by the probe-response pair chain (O-SPEC.PR and O-SPEC.TR discharged);
- (2) it has a known, stable SI frequency $\nu_{\text{SI}}^{\text{cal}}$ supplied by a declared empirical reference process (e.g. a certified atomic or molecular transition);
- (3) it produces a canonical energy difference $\Delta E_{\text{can}}^{\text{cal}}$ in the certified transition ledger; and
- (4) the continuum-frequency bridge (O-SPEC.CI discharged) licenses the identification $\nu_{\text{SPEC}} \leftrightarrow \nu_{\text{SI}}^{\text{cal}}$ without hidden-loss ambiguity.

LEMMA 9.2 ([C]). (Spectroscopic Transition Calibration.) [Conditional on: O-SPEC.PR and O-SPEC.TR discharged; O-SPEC.CI discharged (Thm. 8.9); and a declared calibration-eligible spectroscopic transition T_{cal} (Def. 9.1).] *The certified SPEC transition ledger and continuum-frequency bridge supply a calibration pair*

$$(\Delta E_{\text{can}}^{\text{cal}}, \nu_{\text{SI}}^{\text{cal}}),$$

from which the SI action-unit conversion factor is extracted:

$$\alpha_{\text{act}} = \frac{h_{\text{SI}} \nu_{\text{SI}}^{\text{cal}}}{\Delta E_{\text{can}}^{\text{cal}}}.$$

PROOF. O-SPEC.PR certifies the probe-response pair; O-SPEC.TR certifies the transition ledger; O-SPEC.CI certifies the continuum-frequency identification. The calibration-eligible transition supplies both a canonical energy difference (from the certified transition ledger) and a known SI frequency (from the empirical reference process). By the continuum-frequency bridge (O-SPEC.CI), the correspondence $\nu_{\text{SPEC}} \leftrightarrow \nu_{\text{SI}}^{\text{cal}}$ is licensed without hidden-loss ambiguity. The ratio α_{act} follows from $\Delta E_{\text{SI}} = h_{\text{SI}} \nu_{\text{SI}}$. $\square$

THEOREM 9.3 ([C]). (Planck Constant Calibration via SPEC Transition.) [Conditional on Lem. 9.2; YM action uniqueness (YM Branch, Thm. ??); OS/Wightman reconstruction (YM Branch, O-SPEC.OSW); and a declared calibration-eligible transition T_{cal} .] *The physical Planck constant is the SI calibration of the canonical action unit:*

$$\hbar_{\text{SI}} = \alpha_{\text{act}} \hbar_{\text{can}}.$$

In canonical units, $\hbar_{\text{can}} = 1$. The numerical value $\hbar_{\text{SI}} = 1.054571817 \dots \times 10^{-34} \text{ J} \cdot \text{s}$ enters through the empirical calibration map, not through the foundational derivation.

PROOF. YM action uniqueness (Thm. ??) licenses $\hbar_{\text{can}}$ as the internal quantum action unit (canonical units: $\hbar_{\text{can}} = 1$). The certified SPEC transition ledger and continuum-frequency bridge (O-SPEC.CI) supply the calibration pair $(\Delta E_{\text{can}}, \nu_{\text{SI}})$ via Lem. 9.2. The action-unit conversion factor α_{act} is then the unique positive scale satisfying $\alpha_{\text{act}} \Delta E_{\text{can}} = h_{\text{SI}} \nu_{\text{SI}}$, giving $\hbar_{\text{SI}} = \alpha_{\text{act}} \hbar_{\text{can}}$. $\square$

OPEN OBLIGATION 9.4 ([O]). (O-SPEC.Planck: Planck Calibration — Conditional Frontier.) [Conditional on Thm. 9.3 and a specific declared empirical reference

transition.] Thm. 9.3 is conditional on declaring a calibration-eligible spectroscopic transition T_{cal} . The obligation is to: (1) nominate a specific reference transition (e.g. the caesium hyperfine transition at $\nu_{\text{Cs}} = 9,192,631,770$ Hz, which is the SI definition of the second); (2) certify it as calibration-eligible within the SPEC branch; and (3) verify the canonical energy difference $\Delta E_{\text{can}}^{\text{cal}}$ against the SPEC transition ledger. This is not needed for the Clay YM proof (mathematical units); it is needed if the program claims measured SI physical constants.

REMARK 9.5 ([R]). **Clean separation of roles.** The YM branch licenses $\hbar_{\text{can}} = 1$ (internal canonical action unit, from gauge invariance + locality + renormalizability + positivity forcing the YM action form). The SPEC branch calibrates $\hbar_{\text{SI}}$ (the empirical SI value, from the certified spectroscopic transition ledger and continuum-frequency bridge). The two roles are logically separate: the canonical action unit is a structural theorem; the SI value is a calibration theorem. Neither imports the other as a primitive.

10. SPEC Branch Status Proposition

PROPOSITION 10.1 ([C]). (SPEC Branch Status.) *The SPEC branch currently satisfies:*

- (1) **Lawful branch:** *confirmed by the probe-response realization (Thm. 8.1), source descent from $\mathbf{U}$, Book V legitimacy, and Book VII scoped-certificate discipline.*
- (2) **Gauge-response spectroscopy closure:** *all core obligations $O\text{-SPEC.PR}$ through $O\text{-SPEC.END}$ are conditionally discharged (Thms. 8.1–8.11). The endpoint datum E_{Spec} is fully assembled and ICDC-coherent.*
- (3) **Continuum notation:** *$O\text{-SPEC.CI}$ and $O\text{-SPEC.CD}$ are licensed as Book VI bridges $[B]$ — frequency notation $\nu = \lambda/(2\pi)$ and response-density notation $\rho(\nu) d\nu$ are licensed as effective shorthands.*
- (4) **BLC supplement:** *the SPEC branch carries the Bridge-Ledger Completeness theorem (Thm. 8.6) naming the four-channel no-hidden-loss structure explicitly.*
- (5) **Planck calibration:** *the SPEC branch transition ledger and continuum-frequency bridge supply the canonical downstream location for identifying $\hbar_{\text{SI}}$ via a certified reference transition (Thm. 9.3, conditional).*
- (6) **Status:** *CERT-CLOSE on the gauge-response spectroscopy regime. Not CERT-CLOSE on the matter-rich spectroscopy regime.*
- (7) **Remaining frontiers:**
 - (a) *$O\text{-SPEC.MAT}$ (matter coupling): gated on SM Branch $ob\text{:SM-matter}$ and a lawful coupling theorem.*
 - (b) *$O\text{-SPEC.Planck}$ (Planck calibration): requires declaration of a specific calibration-eligible reference transition.*

PROOF. Items (1)–(4) follow from the discharge section (§8) and the BLC supplement (§8.4). Item (5) from Thm. 9.3. Item (6): gauge-response CERT-CLOSE follows from Thm. 8.11 + ICDC. Item (7): $ob\text{:spec-MAT-open}$ and $ob\text{:SPEC-Planck-calibration}$ state the two remaining frontiers explicitly. $\square$ $\square$